



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name: Mathematics-I
Course Code: TE7680
Faculty: Engineering
Course Credit: 3
Course Level: 1
Sub-Committee (Specialization): Applied Sciences
Learning Objectives:

The students will be able to:

1. Apply successive differentiation to find nth derivative of product of two functions, to evaluate limits of indeterminate forms and to find the series expansion of functions.
2. Apply the concepts of partial differentiation to solve problems on homogeneous functions, Jacobians and maxima & minima.
3. Find the rank of a matrix and solve system of simultaneous equations.
4. Find Eigen values and Eigen vectors, apply Caley-Hamilton theorem to find inverse and higher powers of a matrix and transform matrix to diagonal form.

Books Recommended:

Book	Author	Publisher
Advanced Engineering Mathematics	R. K. Jain and S. R. K. Iyengar	Narosa Publications
Higher Engineering Mathematics	B. S. Grewal	Khanna Publishers
Engineering Mathematics	Ravish R Singh and Mukul Bhatt	McGraw Hill Publications
Engineering Mathematics	N. P. Bali and Manish Goyal	Laxmi Publications
Calculus and Analytical geometry	G. B. Thomas and R. L. Finney	Pearson
A Text Book of Applied Mathematics	P. N. Wartikar & J. N. Wartikar	Pune Vidyarthi GrihaPrakashan
Advanced Engineering Mathematics	Herbert Kreyszig & Erwin Kreyszig	John Wiley & Sons (Asia) Publications

Course Outline:

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Differential Calculus Successive differentiation Leibnitz's theorem, Taylor's and Maclaurin's expansions Indeterminate forms, L'Hospital's rule	7	7
2	Partial Differential Calculus and applications Partial differentiation, Euler's Theorem for homogeneous functions, Total derivatives Jacobians and its properties, Maxima and minima of functions of two variables, Lagrange's method of multipliers	8	8
3	Matrices, vector spaces and sub-spaces Rank of a matrix: Echelon form, Solution of system of equations, Eigenvalues, Eigenvectors Properties of eigenvalues and eigenvectors, Cayley-Hamilton theorem and its applications Euclidian spaces, Vector spaces, Subspaces, Linear Combination, Linearly dependent and independent vectors, Basis, Dimension	15	15

	Change of Basis, Linear transformations, Matrix representation of linear maps. Rank and nullity of a linear transformation, Orthogonality		
4	Tutorial: Tutorial will be conducted on all topics.	15	15
Total		45	45

Pre Requisites:

None

Evaluation:

- A. Continuous Assessment
 - a) Unit Tests
 - b) Tutorials
 - c) Class Tests
 - d) Assignments
- B. End Semester Examination
 - 1. Written examination

Pedagogy:

- 1. Interactive sessions
- 2. Class room teaching
- 3. Tutorials
- 4. Assignment

Expert:

Dr. Arundhati Warke, Professor and Deputy Director Academics, S IT
 Dr. Ramesh Katta, Assistant Professor, SIT